Image statistic encoding between regions of the visual cortex

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Sep 8, 2026

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1 Abstract

Later stages of the visual cortex are thought to contain neurons selective for progressively higher-level visual features. One experimental approach to detecting regional differences in visual processing is through presenting visual texture stimuli (e.g., sand, tree bark, granite, etc.). Families of visual textures can be parameterized by Nth order joint pixel densities, so the decodability of texture families in neural responses can indicate selectivity for image statistics. Previously, Ziemba et al. (2016) found better within-family separation of neural responses in V1 and better between-family separation in V2 [1], and related work found that texture stimuli with matched higher order image statistics produced higher firing in V2 and V4 [2–4]. However, existing analyses lack comparisons with IT, which responds more sparsely to stimuli. Neural responses in IT are sparse in two different ways: first, most IT neurons will not respond to any texture stimuli; and second, of the IT neurons with any response to texture, only a small number of unique stimuli will produce a response. The between-family separability in IT compared to V4 can change depending on the handling of reliable and responsive sites, suggesting that IT efficiently encodes higher-level image statistics.

2 Extended abstract (tentative)

2.1 Higher- and lower-order visual areas may be characterized by differences in processing image statistics

Later stages of the visual cortex are thought to contain neurons selective for progressively higher-level visual features. The primary visual cortex (V1), for instance, contains neurons that selectively respond to simple image features like oriented edges. In contrast, the inferior temporal gyrus (IT), which handles later visual processing, contains areas selective for object categories, such as the fusiform face area for facial recognition.

One approach to characterize the relevant visual features for each layer is through recording neural responses to visual texture stimuli, which are approximately homogeneous images with repeated elements, e.g., images of sand, tree bark, granite, or honeycomb. Texture categories (families) can be parameterized by a set of image statistics, where image statistics are Nth-order empirical densities of pixels.

Unique texture samples within the same family share image statistics and should elicit comparable responses in the visual area sensitive to those particular statistics. Patterns in selectivity for higher- and lower-order image statistics can provide evidence for how visual information is propagated or summarized in visual areas.

2.2 Existing results find a tradeoff between selectivity for higher- and lower-order statistics

Previously, Ziemba et al. (2016) found better within-family separation of neural responses in V1 and better between-family separation in V2 [1]. Relatedly, Okazawa et al. (2015, 2017) compared the selectivity of V2 and V4 neurons for image statistics in texture stimuli and found V4 neurons tended to be selective for higher order statistics [5,6]. Other analyses of between-region texture responses found that texture stimuli with matched higher order image statistics produced higher firing of V2 and V4 neurons compared to stimuli lacking higher order image statistics [2–4].

2.3 It is unknown whether trends in selectivity for image statistics extend beyond V4

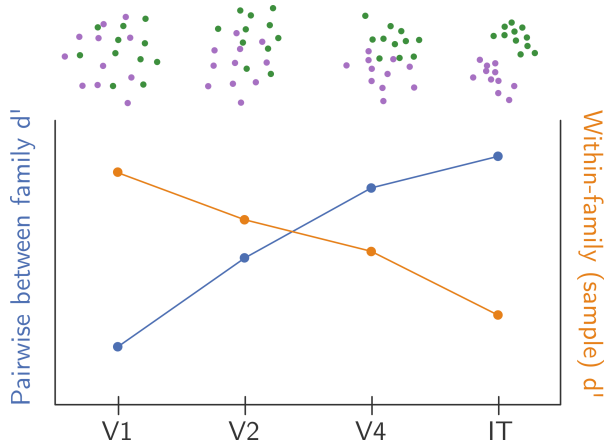


Figure 1: (Top) A visual representation of two families in some arbitrary 2D representation space. (Bottom) The distances between families and between samples, with some arbitrary values for each region.

Lee et al. (2024) recorded neural responses in V1, V2, V4, and IT, spanning the entire range of the visual processing stream [3]. Based on previous results, I expect to find better between-group discrimination in later areas and better within-group discrimination in earlier areas (see Figure 1 for a schematic).

However, existing analyses tend to lack results for the entire visual processing stream, so it is unclear if this trend will reliably extend to IT (Jenelle mentions that previous results conflict over whether texture coding exists in IT but I have not yet filled out this background research). This gap is partially due to the practical limitations of supporting a large population of animals, but there are also methodological challenges with analyzing IT. Higher level areas tend to produce sparse responses, so typical approaches in the literature, such as size-matched population analyses or single-site neuron metrics, may not produce reliable comparisons between

IT and earlier visual areas.

2.4 Results for IT are dependent on the treatment of sparsity

I found that neural responses in IT are sparse in two different ways: first, most IT neurons will not respond to any texture stimuli; and second, of the IT neurons with any response to texture, only a small number of unique stimuli will produce a response. For V1, V2, and V4, recording sites that are unreliable (low correlation between responses to the same stimuli) tend to also be unresponsive (low separability between responses to different stimuli).

Depending on the reliable site criteria, unresponsive sites in IT tend to achieve above-threshold reliability. Retaining reliable sites in IT produces worse between-family separability in IT compared to V4, while limiting analyses to responsive sites reverses this trend, producing better between-family separability in IT compared to V4. The within-family separability in IT remains consistently worse than V4. This contrast suggests that IT destructively summarizes pixel-level differences in images but efficiently encodes higher-level image statistics.

3 Timeline

3.1 Milestones

A rough draft is due on October 9th. A final report is due on December 9th. The ADA presentation usually happens in November or December, though some presentations may occur in the spring.

3.2 Week-by-week

1. (Sep 07) When reporting Mahalanobis distance between families or between samples, I want to characterize how the trends change depending on the inclusion of reliable sites when the criteria is based on split-half correlation when aggregated over family or aggregated over image (tentative results already available).
2. (Sep 14) Currently, my results depend on subsampling neurons. Thus, it would be reasonable to either scale the results to multiple bootstrapped subsamples or to explore other methods that are less reliant on dimensionality. There don't seem to be many options for te latter, so it would probably be more useful to focus on the bootstrapping pipeline and put in other metrics later.
3. (Sep 21) I expect the previous goal to take two weeks. Especially since I will likely need to run the permutation nulls and other control experiments on the non-naturalistic textures.
4. (Sep 28) Based on my results with reliable sites, I want to further specify how sparsity affects the results for IT (and to a lesser extent V1, which is "sparse" in the sense that one array seems faulty).
5. (Oct 05) Writing up results; the first draft should be complete on October 9th.
6. (Oct 12) Fall break.
7. (Oct 19) Probably continuation of the goal from (Sep 28).
8. (Oct 26) Concrete steps will likely depend on the pre-Fall break results. Likely running the permutations necessary for a null distribution of relevant metrics.
9. (Nov 02) Concrete steps will likely depend on the pre-Fall break results.
10. (Nov 09) No additional analyses should be completed beyond this point, as time should be dedicated toward the presentation and refining the final report.
11. (Nov 16) Possible presentation.
12. (Nov 23) Possible presentation.
13. (Nov 30) Possible presentation.
14. (Dec 7) The final report is due on December 9th.

4 Tentative results and commentary

4.1 Characterizing texture information

The degree of "texture encoding information" in IT is a bit difficult to fully characterize with the given data since we lack the real texture statistics of the stimuli. As a proxy, we can consider whether neural responses contain encoding information about the texture family.

Mahalanobis distance can capture linear separability but is not robust to variation in dimensionality. Generalization performance of regularized regression models can be applied to between-family but not within-family comparisons (due to lack of repetitions). Mutual information measures are also dependent on dimensionality but may produce a more general metric of dependence rather than assuming a geometric structure to the separation.

4.2 Site selection

Doing site selection prior to analysis isn't necessarily bad. Considering sites are typically not i.i.d., it might be useful to discard sites based on a known criteria rather than use all the sites in every region. Retaining all sites then correcting based on the different dimensions could be misleading because of the non-i.i.d. nature of the sites.

However, site selection does introduce biases. Previously, I computed reliability with split-half correlation after aggregating over the texture family then splitting by repetition number (even in one group, odd in the other, for example). This strategy is consistent with Gerick's previous work but also introduces the pattern of reliable IT sites being unresponsive. Instead, one could do reliability measures with split-half correlation aggregated over the sample image.

In either case, the distributions of split-half correlations are markedly different between regions (Figure 2, though the family aggregation results not shown). For example, the proportion of sites with a split-half correlation below 0.25 is approximately 0.6 for V1, 0.15 for V2, 0.4 for V4, and 0.75 for IT.

4.3 Matched-reliability results between families

Jenelle suggests "reliability-matching" regions. For example, instead of sub-sampling from all sites above a threshold, I would sub-sample sites with split-half correlations in a similar range. This produces Figure 3, which shows a trend of increasing Mahalanobis distances between families in progressive visual areas.

Previously, when doing reliable site selection with Gerick's family-aggregated method, IT distance falls *below* V4.

I basically agree with "reliability-matching" as a decent way to address "sparsity" concerns. It discards the noisy sites and also prevents the higher proportion of really reliable sites in V2

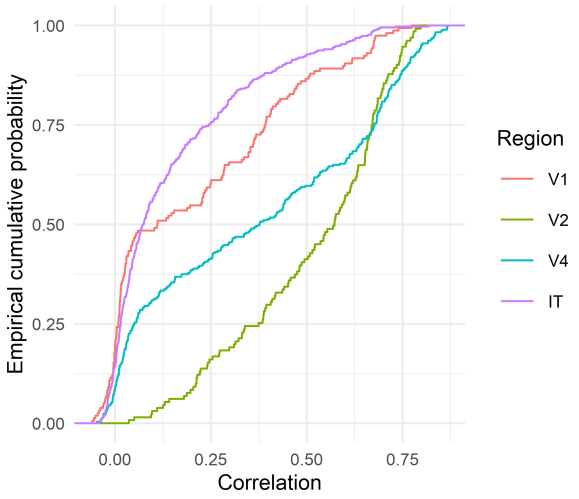


Figure 2: Empirical cumulative distribution functions for the split-half correlations by region. Data reflects split-half correlations performed when grouped by sample images Around 75 percent of IT sites fall below a split-half correlation of 0.25.

and V4 from basically dictating the distance/information metrics.

I did a hack job with Figure 3, where I chose a range of split-half correlation between 0.3 and 0.6 since the eCDFs in Figure 2 look to have a similar slope in that range. There are some modifications to just subsampling in a range. E.g., I could use the empirical CDF to correct the probability of sampling any given site such that the split-half correlations in the subsampled sites are approximately uniform.

4.4 Between-sample results

I am still running the calculations for this, as I'm using Jenelle's suggestion of randomized dichotomy calculations (essentially manifold capacity). Within a family, samples are separated into two groups of 7 and the distance calculated between them (see previous reports for more details).

Pairwise between-sample comparisons could also work. Previous results suggest that these are pretty dependent on site reliability (though I have not dived into this very deeply).

4.5 Sparsity reports

Because of the non-i.i.d. site sampling, I would want to avoid claims about the actual proportion of sparsity in IT. Other criteria, like the number of images a site is responsive to given that a site is responsive to at least one image, provide a more generic measure.

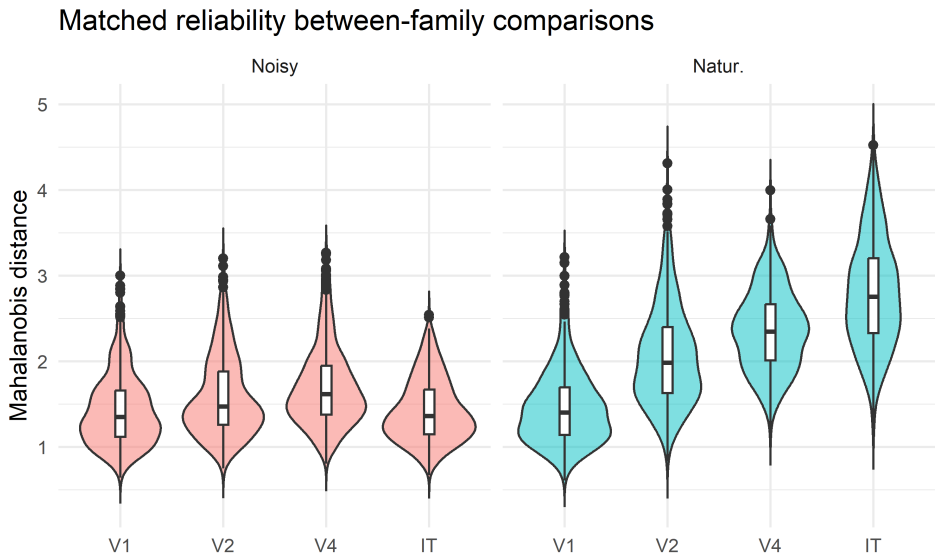


Figure 3: Mahalanobis distances between texture families in subsampled sites with split-half correlations between 0.3 and 0.6. In naturalistic textures (right), the distance between families increases with progressive visual areas. The noisy textures, which lack coherent texture statistics, serve as a sanity check (left).

Sparsity is likely the cause of the difference in reliable/responsive site calculations for Gerick's method of site selection (aggregated over family) and my current method (aggregated over specific image).

5 Discussion

Site selection based on matched reliability (split-half correlation in a similar range when entries are aggregated over specific image and split by repetition number) is intended to control for the sensitivity of distance/information measures to dimensionality by discarding noisy sites in each region. As currently implemented, I would not be able to answer questions like "what's the theoretical ceiling of texture information between regions?" since it discards both very reliable and very unreliable sites. However, I suspect questions of this type are a bit impossible to answer anyway considering the limited number of sites.

Texture families are a limited proxy of the target measure of image statistic selectivity. I currently have no information about the baseline similarity of families to each other. I am pretty sure (based on EDA) that there are some families that produce reliable neural responses in every region. It would be good to see if the results hold in those families specifically.

Multi-unit activations may record from neighboring neurons, so any discussion of sparsity should be attentive to the fact that specificity is obscured (mostly relevant for IT).

Between-region and between-subject sparsity considerations prevents easy selection of a hold-out set.

6 References

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